

Inverse Asymptotics for a Real-Argument Fibonacci Function

Log-periodic transseries, closed all-order coefficients, and a general multiexponential reversion calculus

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Abstract

We study the real interpolation

$$\mathcal{F}(x) = \sqrt{\frac{2}{5}} \sqrt{\cosh(2x \operatorname{arccsch} 2) - \cos(\pi x)}, \quad x \in \mathbb{R},$$

which satisfies $\mathcal{F}(n) = \operatorname{Fib}_n$ for every nonnegative integer n . Because $\mathcal{F}$ is even, every sufficiently large value y has two real preimages, $X_{\pm}(y) = \pm X(y)$. The dominant inverse is $L = \log_{\varphi}(\sqrt{5}y)$, but the correction is not an ordinary power series with constant coefficients. It is a convergent log-periodic transseries. With

$$q = (5y^2)^{-1}, \quad \theta = \pi L, \quad \tau = \frac{\pi}{2 \log \varphi},$$

we prove

$$X(y) = L + \frac{1}{2 \log \varphi} \sum_{N \geq 1} q^N P_N(\theta; \tau),$$

where each P_N is an explicit real trigonometric polynomial containing only harmonics of the same parity as N . More strongly, after introducing two conjugate complex monomials, every coefficient is available in one closed formula involving generalized binomial coefficients. The series is the actual Taylor expansion of a holomorphic implicit solution near the transseries origin, so total-degree truncation after N blocks has a uniform $\mathcal{O}(y^{-2N-2})$ remainder.

The derivation leads to a general inversion calculus for functions of the form

$$Y(x) = C e^{\alpha x} \Psi(\zeta_1 e^{-\mu_1 x}, \dots, \zeta_m e^{-\mu_m x}), \quad \Re \mu_j > 0, \quad \Psi(\mathbf{0}) = 1.$$

For the inverse, the coefficient indexed by a multi-index $\mathbf{n} \neq \mathbf{0}$ is a single diagonal coefficient of the variable power $\Psi^{\Gamma_{\mathbf{n}}/\alpha}$, where $\Gamma_{\mathbf{n}} = \mathbf{n} \cdot \boldsymbol{\mu}$. We develop the underlying Lagrange–Bürmann identity in filtered transseries algebras, its diagonal exponential coefficient formula, periodic and quasiperiodic block versions, conjugation by a nonlinear dominant skeleton, symbolic algorithms, residual certificates, and numerical validation.

Repository context. This note is designed as a standalone article for the series-and-transseries research area of the ProveIt repository:

<https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/series-and-transseries>.

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1 Problem, notation, and principal result

Let

$$\lambda := \operatorname{arccsch} 2 = \operatorname{arsinh} \frac{1}{2} = \log \frac{1 + \sqrt{5}}{2} = \log \varphi, \quad a := 2\lambda, \quad b := \pi, \quad (1.1)$$

where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio. We use

$$\mathcal{F}(x) := \sqrt{\frac{2}{5}} \sqrt{\cosh(ax) - \cos(bx)}. \quad (1.2)$$

The square root is the nonnegative real square root. The quantities used throughout the inverse expansion are

$$\begin{aligned} R &:= \sqrt{5}y, & L &:= \frac{\log R}{\lambda} = \log_{\varphi} R, \\ q &:= R^{-2} = \frac{1}{5y^2} = e^{-aL}, & \theta &:= bL, & \tau &:= \frac{b}{a} = \frac{\pi}{2\lambda}. \end{aligned} \quad (1.3)$$

Numerically,

$$\lambda = 0.48121182505960344749\dots, \quad \tau = 3.26425130263649690673\dots \quad (1.4)$$

The dominant estimate $\mathcal{F}(x) \sim \varphi^x / \sqrt{5}$ gives $x \sim L$. The oscillatory term $\cos(\pi x)$, however, survives inversion as a periodic function of $\log y$. It is therefore essential to retain the phase $\theta = \pi \log_{\varphi}(\sqrt{5}y)$.

For the closed formula introduce the conjugate monomials

$$\xi := e^{-(a+ib)L} = qe^{-i\theta} = R^{-2-i\pi/\lambda}, \quad \eta := e^{-(a-ib)L} = qe^{i\theta} = R^{-2+i\pi/\lambda}, \quad (1.5)$$

and, for $r, s \in \mathbb{N}$,

$$\chi_{r,s} := r + s + i\tau(r - s). \quad (1.6)$$

Here and below $\mathbb{N} = \{0, 1, 2, \dots\}$, and for $n \in \mathbb{N}$, $\binom{z}{n} = z(z-1)\cdots(z-n+1)/n!$ with $\binom{z}{0} = 1$.

Principal inverse-transseries formula

For every real $y \geq 2$, the equation $\mathcal{F}(x) = y$ has exactly two real solutions $X_{\pm}(y) = \pm X(y)$, with $X(y) > 0$. The positive branch has the convergent expansion

$$X(y) = L + \sum_{\substack{r,s \geq 0 \\ r+s \geq 1}} C_{r,s} \xi^r \eta^s, \quad C_{r,s} = \frac{(-1)^{r+s+1}}{a\chi_{r,s}} \binom{\chi_{r,s}}{r} \binom{\chi_{r,s}}{s}. \quad (1.7)$$

The coefficients obey $C_{s,r} = \overline{C_{r,s}}$, hence the value on the physical ray $\eta = \bar{\xi}$ is real. Grouping terms of total degree $N = r + s$ gives

$$X(y) = L + \frac{1}{a} \sum_{N \geq 1} q^N P_N(\theta; \tau), \quad (1.8)$$

where

$$P_N(\theta; \tau) := \sum_{r=0}^N \beta_{r,N-r} e^{i(N-2r)\theta}, \quad (1.9)$$

$$\beta_{r,s} := aC_{r,s} = \frac{(-1)^{r+s+1}}{\chi_{r,s}} \binom{\chi_{r,s}}{r} \binom{\chi_{r,s}}{s}. \quad (1.10)$$

For each fixed N , P_N is a real 2π -periodic trigonometric polynomial with harmonics $N, N - 2, \dots, -N$. For every fixed truncation order N ,

$$X(y) = L + \sum_{\substack{r,s \geq 0 \\ 1 \leq r+s \leq N}} C_{r,s} \xi^r \eta^s + \mathcal{O}(q^{N+1}) = L + \frac{1}{a} \sum_{n=1}^N q^n P_n(\theta; \tau) + \mathcal{O}(y^{-2N-2}), \quad (1.11)$$

uniformly in the phase as $y \rightarrow +\infty$. The negative branch is obtained without further calculation:

$$X_-(y) = -X(y). \quad (1.12)$$

The first four blocks are

$$P_1 = 2 \cos \theta, \quad (1.13)$$

$$P_2 = -2 - \cos(2\theta) - 2\tau \sin(2\theta), \quad (1.14)$$

$$P_3 = (6 - \tau^2) \cos \theta + 5\tau \sin \theta + \frac{2 - 9\tau^2}{3} \cos(3\theta) + 3\tau \sin(3\theta), \quad (1.15)$$

$$P_4 = -9 + 4(3\tau^2 - 2) \cos(2\theta) + \frac{4\tau(2\tau^2 - 13)}{3} \sin(2\theta) + \frac{16\tau^2 - 1}{2} \cos(4\theta) + \frac{\tau(16\tau^2 - 11)}{3} \sin(4\theta). \quad (1.16)$$

Thus, in powers of y^{-2} ,

$$X(y) = L + \frac{\cos \theta}{5\lambda y^2} - \frac{2 + \cos(2\theta) + (\pi/\lambda) \sin(2\theta)}{50\lambda y^4} + \frac{1}{250\lambda y^6} \left[(6 - \tau^2) \cos \theta + 5\tau \sin \theta + \frac{2 - 9\tau^2}{3} \cos(3\theta) + 3\tau \sin(3\theta) \right] + \mathcal{O}(y^{-8}). \quad (1.17)$$

The periodic coefficients are not an ornamental rewriting: as shown in [section 6](#), no constant-coefficient Poincaré series in integer powers of y^{-2} can represent this inverse.

2 The real Fibonacci interpolation and its inverse branches

2.1 Interpolation of the Fibonacci sequence

The defining expression admits the useful modulus form

$$5\mathcal{F}(x)^2 = e^{2\lambda x} + e^{-2\lambda x} - 2 \cos(\pi x) = \left| e^{\lambda x} - e^{(-\lambda + i\pi)x} \right|^2. \quad (2.1)$$

For an integer $n \geq 0$, $e^{i\pi n} = (-1)^n$, and therefore

$$\begin{aligned} 5\mathcal{F}(n)^2 &= \varphi^{2n} + \varphi^{-2n} - 2(-1)^n \\ &= (\varphi^n - (-\varphi^{-1})^n)^2 = 5 \text{Fib}_n^2. \end{aligned} \quad (2.2)$$

Both sides are nonnegative, so $\mathcal{F}(n) = \text{Fib}_n$. The function is even and strictly positive away from the origin. Near zero,

$$\mathcal{F}(x) = \sqrt{\frac{a^2 + b^2}{5}} |x| + \mathcal{O}(|x|^3), \quad (2.3)$$

so the even interpolation has a cusp at $x = 0$, despite the analyticity of $\mathcal{F}(x)^2$.

2.2 Why only two real branches survive at large height

Put

$$h(x) := \cosh(ax) - \cos(bx), \quad \mathcal{F}(x)^2 = \frac{2}{5}h(x). \quad (2.4)$$

For $x > 0$,

$$h'(x) = a \sinh(ax) + b \sin(bx). \quad (2.5)$$

Let

$$x_* := \frac{1}{a} \operatorname{arsinh} \frac{b}{a}. \quad (2.6)$$

For $x > x_*$, one has $a \sinh(ax) > b$, hence $h'(x) > b + b \sin(bx) \geq 0$, with strict positivity. Define

$$M := \max_{0 \leq x \leq x_*} \mathcal{F}(x). \quad (2.7)$$

Every $y > M$ has no positive preimage in $[0, x_*]$, while the restriction of $\mathcal{F}$ to (x_*, ∞) is strictly increasing and unbounded. It has therefore one positive preimage, and evenness supplies exactly one negative preimage.

An entirely explicit, if nonoptimal, sufficient height follows from $\cos(bx) \geq -1$:

$$M \leq \sqrt{\frac{2}{5} (\cosh(ax_*) + 1)} = \sqrt{\frac{2}{5} \left(\sqrt{1 + (b/a)^2} + 1 \right)} = 1.3287575092 \dots \quad (2.8)$$

For orientation, numerical refinement gives $x_* = 1.9729927341 \dots$, and the actual maximum on $[0, x_*]$ is $1.0138598855 \dots$, attained near $x = 1.1323790680 \dots$. None of the asymptotic analysis depends on this numerical refinement.

Remark 2.1 (Low-height inverse branches). *The oscillatory term can create several local inverse branches for bounded values of y . Those branches terminate at finite critical values and play no role in the limit $y \rightarrow \infty$. Throughout the rest of the article, $X(y)$ denotes the unique positive outer branch.*

3 Exact reduction to near-identity reversion

The transseries follows from an exact factorization rather than from repeated ad hoc substitution. With

$$\mu_+ := a + ib, \quad \mu_- := a - ib, \quad (3.1)$$

we have

$$\begin{aligned} R^2 = 5y^2 &= e^{ax} + e^{-ax} - 2 \cos(bx) \\ &= e^{ax} (1 - e^{-\mu_+ x}) (1 - e^{-\mu_- x}). \end{aligned} \quad (3.2)$$

For real $x > 0$, the two factors are conjugate, and each lies in the open right half-plane because $|e^{-\mu_{\pm} x}| < 1$. Principal logarithms are therefore unambiguous. Taking the positive square root and then the logarithm gives

$$\log R = \lambda x + \frac{1}{2} \log (1 - e^{-\mu_+ x}) + \frac{1}{2} \log (1 - e^{-\mu_- x}). \quad (3.3)$$

After division by $\lambda = a/2$, the inverse problem is exactly

$$L = H(x) := x + A(x), \quad (3.4)$$

where

$$A(x) := \frac{1}{a} [\log(1 - e^{-\mu+x}) + \log(1 - e^{-\mu-x})] \quad (3.5)$$

$$= \frac{1}{a} \log(1 - 2e^{-ax} \cos(bx) + e^{-2ax}) \quad (3.6)$$

$$= -\frac{2}{a} \sum_{m \geq 1} \frac{e^{-amx} \cos(mbx)}{m}. \quad (3.7)$$

The last series is absolutely convergent for every $x > 0$.

Set

$$X(y) = L + U(L). \quad (3.8)$$

Then U is characterized by the exact implicit equation

$$U + \frac{1}{a} [\log(1 - \xi e^{-\mu+U}) + \log(1 - \eta e^{-\mu-U})] = 0, \quad (3.9)$$

where ξ, η are given by [equation \(1.5\)](#). There is also a useful purely real form. Dividing the original squared equation by $e^{aL} = R^2$ yields

$$e^{aU} + q^2 e^{-aU} - 2q \cos(\theta + bU) = 1. \quad (3.10)$$

If

$$W := aU = \sum_{N \geq 1} q^N P_N(\theta; \tau), \quad (3.11)$$

then [equation \(3.10\)](#) becomes the compact block-generating equation

$$\boxed{e^W + q^2 e^{-W} - 2q \cos(\theta + \tau W) = 1.} \quad (3.12)$$

This identity is an independent way to generate and verify every polynomial P_N .

4 Closed all-order coefficients

4.1 A one-variable reversion identity

We begin with the classical near-identity form of the Lagrange–Bürmann formula [\[1, 2\]](#). Let $D = d/dX$.

Theorem 4.1 (Near-identity Lagrange–Bürmann formula). *Suppose*

$$H(X) = X + A(X) \quad (4.1)$$

has a local inverse $H^{-1}(X) = X + B(X)$, with A small in an analytic or complete filtered formal sense. Then

$$\boxed{B(X) = \sum_{n \geq 1} \frac{(-1)^n}{n!} D^{n-1}(A(X)^n).} \quad (4.2)$$

The first terms are

$$B = -A + AA' - A(A')^2 - \frac{1}{2}A^2A'' + \dots. \quad (4.3)$$

Proof. Introduce a bookkeeping parameter t and solve

$$B_t = -tA(X + B_t), \quad B_0 = 0. \quad (4.4)$$

The standard Lagrange inversion formula for $w = t\Phi(X + w)$ gives

$$w = \sum_{n \geq 1} \frac{t^n}{n!} D^{n-1}(\Phi(X)^n). \quad (4.5)$$

Taking $\Phi = -A$ and then setting $t = 1$ gives equation (4.2). Expansion through total degree three in A and its derivatives gives equation (4.3). $\square$

For an exponential transseries, differentiation is diagonal on each monomial. This turns equation (4.2) into a particularly economical coefficient formula.

Proposition 4.2 (Diagonal exponential coefficient formula). *Let*

$$A(X) = \sum_{\gamma \in \mathfrak{G}_+} a_\gamma e^{-\gamma X}, \quad \Re \gamma > 0, \quad (4.6)$$

where the support is locally finite under addition. Write

$$B(X) = \sum_{\gamma \in \mathfrak{G}_+} b_\gamma e^{-\gamma X}. \quad (4.7)$$

Then, for every nonzero exponent γ ,

$$b_\gamma = - \sum_{n \geq 1} \frac{\gamma^{n-1}}{n!} [e^{-\gamma X}] A(X)^n \quad (4.8)$$

$$= - \frac{1}{\gamma} [e^{-\gamma X}] \exp(\gamma A(X)). \quad (4.9)$$

Only finitely many values of n contribute to any fixed coefficient in a positively filtered grid.

Proof. On the monomial $e^{-\gamma X}$,

$$D^{n-1} e^{-\gamma X} = (-\gamma)^{n-1} e^{-\gamma X}. \quad (4.10)$$

Consequently the coefficient contributed by the n -th summand of equation (4.2) is

$$\frac{(-1)^n}{n!} (-\gamma)^{n-1} [e^{-\gamma X}] A^n = - \frac{\gamma^{n-1}}{n!} [e^{-\gamma X}] A^n. \quad (4.11)$$

Summing over n proves both forms. $\square$

Remark 4.3 (Why the formula is “diagonal”). *For each target exponent γ , one raises the entire perturbation $\exp(A)$ to the scalar power γ , extracts only the coefficient at that same exponent, and divides by $-\gamma$. No recursive substitution is needed. The dependence of the power on the coefficient being extracted is the feature that packages the inversion combinatorics.*

4.2 Application to the Fibonacci interpolation

At $X = L$, equation (3.5) reads

$$A(L) = \frac{1}{a} (\log(1 - \xi) + \log(1 - \eta)). \quad (4.12)$$

The derivative $D = d/dL$ acts by

$$D\xi = -\mu_+\xi, \quad D\eta = -\mu_-\eta. \quad (4.13)$$

Thus the monomial $\xi^r \eta^s$ has exponent

$$\Gamma_{r,s} := r\mu_+ + s\mu_- = a\chi_{r,s}. \quad (4.14)$$

By theorem 4.2, its coefficient in the inverse correction is

$$\begin{aligned} C_{r,s} &= -\frac{1}{\Gamma_{r,s}} [\xi^r \eta^s] \exp(\Gamma_{r,s} A(L)) \\ &= -\frac{1}{a\chi_{r,s}} [\xi^r \eta^s] (1 - \xi)^{\chi_{r,s}} (1 - \eta)^{\chi_{r,s}} \\ &= \frac{(-1)^{r+s+1}}{a\chi_{r,s}} \binom{\chi_{r,s}}{r} \binom{\chi_{r,s}}{s}. \end{aligned} \quad (4.15)$$

This proves the coefficient formula in equation (1.7). Notice that $\Re \chi_{r,s} = r + s > 0$, so its denominator never vanishes for $r + s \geq 1$.

On the physical ray, $\eta = \bar{\xi}$. Since

$$\chi_{s,r} = \overline{\chi_{r,s}}, \quad C_{s,r} = \overline{C_{r,s}}, \quad (4.16)$$

terms pair to give a real result. Moreover,

$$\xi^r \eta^s = q^{r+s} e^{i(s-r)\theta}. \quad (4.17)$$

Collecting all pairs with $r + s = N$ gives equations (1.8) and (1.9). The available frequencies are $s - r = N - 2r$, which proves the parity statement for the harmonics.

An alternative single-line version of the expansion, directly in terms of complex powers of the large argument, is

$$X(y) = \frac{\log(\sqrt{5}y)}{\lambda} + \sum_{\substack{r,s \geq 0 \\ r+s \geq 1}} C_{r,s} (\sqrt{5}y)^{-2(r+s) - i(\pi/\lambda)(r-s)}. \quad (4.18)$$

Although individual terms are complex, conjugate pairing makes the sum real. This is the natural generalized-power transseries hidden behind the real log-periodic blocks.

5 The log-periodic blocks in detail

5.1 Generation and parity

The real logarithmic perturbation at the dominant inverse is

$$A(L) = -\frac{2}{a} \sum_{m \geq 1} \frac{q^m \cos(m\theta)}{m}. \quad (5.1)$$

In the (q, θ) representation, differentiation with respect to L is

$$D = -aq \frac{\partial}{\partial q} + b \frac{\partial}{\partial \theta}. \quad (5.2)$$

Substitution of equation (5.1) into equation (4.2), followed by collection of powers of q , is a generic way to construct the blocks. For this particular problem, the binomial formula equation (1.10) is faster and exposes their exact arithmetic structure. The implicit generating equation equation (3.12) gives a third, independent construction.

For example, the total-degree-one pairs $(r, s) = (1, 0), (0, 1)$ each have $\beta = 1$, giving $P_1 = 2 \cos \theta$. At total degree two, the central pair $(1, 1)$ contributes -2 , while the conjugate edge pair contributes $-\cos(2\theta) - 2\tau \sin(2\theta)$. This recovers equation (1.14).

Proposition 5.1 (Block symmetries). *For every $N \geq 1$,*

$$P_N(\theta + 2\pi; \tau) = P_N(\theta; \tau), \quad (5.3)$$

$$P_N(\theta + \pi; \tau) = (-1)^N P_N(\theta; \tau), \quad (5.4)$$

$$P_N(-\theta; -\tau) = P_N(\theta; \tau). \quad (5.5)$$

If N is odd, P_N has zero phase average. If $N = 2m$, then

$$\frac{1}{2\pi} \int_0^{2\pi} P_{2m}(\theta; \tau) d\theta = -\frac{1}{2m} \binom{2m}{m}^2. \quad (5.6)$$

Proof. The first two identities follow from the frequency set $N - 2r$. The third follows by complex conjugation. A constant Fourier term can occur only when $r = s = m$, hence only for $N = 2m$. In that case $\chi_{m,m} = 2m$, and equation (1.10) gives

$$\beta_{m,m} = -\frac{1}{2m} \binom{2m}{m}^2. \quad (5.7)$$

□

Consequently the phase-averaged inverse correction has the exact series

$$\langle X - L \rangle_\theta = -\frac{1}{a} \sum_{m \geq 1} \frac{1}{2m} \binom{2m}{m}^2 q^{2m}. \quad (5.8)$$

Using

$${}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; u\right) = \sum_{m \geq 0} \binom{2m}{m}^2 \frac{u^m}{16^m}, \quad (5.9)$$

this can also be written as

$$\langle X - L \rangle_\theta = -\frac{1}{2a} \int_0^{16q^2} \frac{{}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; u) - 1}{u} du. \quad (5.10)$$

The mean is negative, even though the leading pointwise correction $q \cos \theta / \lambda$ changes sign.

5.2 Multiplicative phase recurrence

Because $\theta = \pi \log_\varphi(\sqrt{5}y)$, multiplying y by φ^2 adds 2π to the phase:

$$\theta(\varphi^2 y) = \theta(y) + 2\pi, \quad q(\varphi^2 y) = \varphi^{-4} q(y). \quad (5.11)$$

Thus the coefficient profiles repeat on the multiplicative scale $y \mapsto \varphi^2 y$, while their amplitudes contract geometrically. Multiplication by φ adds π to the phase and, by equation (5.4), changes the sign of every odd block. This discrete scale covariance is the real-variable manifestation of the complex exponents in equation (4.18).

5.3 Scaled implicit equation as a consistency check

Writing $W = \sum_{N \geq 1} q^N P_N$, expand [equation \(3.12\)](#) as a formal power series in q . The coefficient of q gives

$$P_1 - 2 \cos \theta = 0. \quad (5.12)$$

At order q^2 , one obtains

$$P_2 + \frac{1}{2} P_1^2 + 1 + 2\tau P_1 \sin \theta = 0, \quad (5.13)$$

which simplifies to [equation \(1.14\)](#). Every higher block is determined linearly after the preceding blocks are known; this makes [equation \(3.12\)](#) particularly effective as a symbolic residual test.

6 Analytic status, remainders, and the necessity of transseries

6.1 The expansion is convergent near the transseries origin

The word *transseries* is sometimes used only for divergent formal objects; standard background on asymptotic expansions and transseries may be found in [\[4, 5, 6, 7\]](#). In the present problem the generalized expansion is stronger: it is the Taylor series of a holomorphic function of two independent small variables.

Define

$$\mathcal{G}(U, z, w) := U + \frac{1}{a} [\log(1 - ze^{-\mu+U}) + \log(1 - we^{-\mu-U})]. \quad (6.1)$$

Near $(U, z, w) = (0, 0, 0)$, this is holomorphic and satisfies

$$\mathcal{G}(0, 0, 0) = 0, \quad \frac{\partial \mathcal{G}}{\partial U}(0, 0, 0) = 1. \quad (6.2)$$

The holomorphic implicit-function theorem therefore gives a unique convergent power series

$$U(z, w) = \sum_{r+s \geq 1} C_{r,s} z^r w^s \quad (6.3)$$

in some bidisc $|z| < \rho, |w| < \rho$. Formal uniqueness and [equation \(4.15\)](#) identify its Taylor coefficients with the closed formula already obtained.

On the physical ray $(z, w) = (\xi, \eta)$,

$$|z| = |w| = q = \frac{1}{5y^2} \longrightarrow 0. \quad (6.4)$$

Hence [equation \(1.7\)](#) converges absolutely for all sufficiently large y . Cauchy estimates and coefficient extraction in a smaller bidisc (compare [\[3\]](#)) give, for every fixed N ,

$$\sum_{r+s \geq N+1} C_{r,s} \xi^r \eta^s = \mathcal{O}(q^{N+1}), \quad (6.5)$$

with a constant independent of $\arg \xi = -\theta$. This proves the uniform remainder claim [equation \(1.11\)](#). The same reasoning permits termwise differentiation with respect to L , and therefore with respect to y , inside any smaller convergence domain.

The local argument can be supplemented by an explicit sufficient convergence region.

Proposition 6.1 (A concrete bidisc of convergence). *Let*

$$m := |\mu_{\pm}| = \sqrt{a^2 + b^2}, \quad k := \frac{2m}{a} = 2\sqrt{1 + \tau^2}, \quad (6.6)$$

and define

$$q_{\text{suf}} := \frac{k^k}{(k+1)^{k+1}} = 0.05024103075902923436 \dots \quad (6.7)$$

The Taylor series equation (6.3) converges throughout

$$\max(|z|, |w|) < q_{\text{suf}}. \quad (6.8)$$

Consequently, on the physical ray it converges whenever

$$y > \frac{1}{\sqrt{5}q_{\text{suf}}} = 1.9951967438470964 \dots, \quad (6.9)$$

and in particular for every real $y \geq 2$.

Proof. Rewrite equation (3.9) as $U = \Phi(U; z, w)$, where

$$\Phi(U; z, w) := -\frac{1}{a} [\log(1 - ze^{-\mu+U}) + \log(1 - we^{-\mu-U})]. \quad (6.10)$$

Choose $0 < \rho < 1/(k+1)$ and set

$$R_{\rho} := -\frac{2}{a} \log(1 - \rho), \quad q_{\rho} := \rho e^{-mR_{\rho}} = \rho(1 - \rho)^k. \quad (6.11)$$

If $|U| \leq R_{\rho}$ and $|z|, |w| \leq q_{\rho}$, then $|ze^{-\mu+U}|, |we^{-\mu-U}| \leq \rho$. The elementary bound $|\log(1 - v)| \leq -\log(1 - |v|)$ gives

$$|\Phi(U; z, w)| \leq -\frac{2}{a} \log(1 - \rho) = R_{\rho}. \quad (6.12)$$

Moreover,

$$|\partial_U \Phi| \leq \frac{2m}{a} \frac{\rho}{1 - \rho} = k \frac{\rho}{1 - \rho} < 1. \quad (6.13)$$

Thus Φ is a uniform contraction of the closed disk $|U| \leq R_{\rho}$, and its fixed point depends holomorphically on (z, w) . The function $q_{\rho} = \rho(1 - \rho)^k$ increases up to the endpoint $\rho = 1/(k+1)$, where its supremum is $k^k/(k+1)^{k+1}$. Every smaller bidisc is therefore covered by some strict contraction, proving the claim. $\square$

Definition 6.2 (Block-Poincaré expansion). *An expansion*

$$f(y) \sim \sum_{N \geq 0} q(y)^N P_N(\theta(y)) \quad (6.14)$$

is called a block-Poincaré expansion if every P_N is bounded on its phase domain and truncation after N has remainder $\mathcal{O}(q^{N+1})$. Individual blocks need not form a strict pointwise asymptotic scale, because a periodic coefficient may vanish at isolated phases.

The expansion equation (1.8) is a convergent block-Poincaré expansion. It is also a grid-based exponential transseries in L , and a generalized complex-power series in $R = \sqrt{5}y$.

6.2 Why an ordinary power series cannot work

The leading correction is

$$X(y) - L = \frac{q}{\lambda} \cos \theta + \mathcal{O}(q^2). \quad (6.15)$$

Consider the two sequences

$$y_n^{(+)} := \frac{\varphi^{2n}}{\sqrt{5}}, \quad y_n^{(-)} := \frac{\varphi^{2n+1}}{\sqrt{5}}. \quad (6.16)$$

Along the first, $\theta = 2\pi n$; along the second, $\theta = (2n + 1)\pi$. Therefore

$$\lim_{n \rightarrow \infty} \frac{X(y_n^{(+)}) - L(y_n^{(+)})}{q(y_n^{(+)})} = \frac{1}{\lambda}, \quad \lim_{n \rightarrow \infty} \frac{X(y_n^{(-)}) - L(y_n^{(-)})}{q(y_n^{(-)})} = -\frac{1}{\lambda}. \quad (6.17)$$

There is no single first constant coefficient. In particular, no expansion of the form

$$X(y) = L + c_1 y^{-2} + c_2 y^{-4} + \dots \quad (6.18)$$

can be asymptotic in the usual constant-coefficient sense. The logarithmic phase, or equivalently the pair of complex powers, is mathematically forced.

6.3 Residuals give a posteriori error certificates

Let $B_N(L)$ be any approximation to $U(L)$, and define the exact residual for the near-identity equation by

$$\mathcal{R}_N(L) := H(L + B_N(L)) - L = B_N(L) + A(L + B_N(L)). \quad (6.19)$$

If $|A'(x)| \leq \kappa < 1$ on the real segment joining $L + B_N(L)$ and $L + U(L)$, the mean-value theorem gives

$$|U(L) - B_N(L)| \leq \frac{|\mathcal{R}_N(L)|}{1 - \kappa}. \quad (6.20)$$

Thus a small residual is not merely a numerical diagnostic; once a derivative bound is supplied it certifies the inverse error. For the exact transseries truncation through total degree N , formal substitution gives

$$\mathcal{R}_N(L) = \mathcal{O}(q^{N+1}). \quad (6.21)$$

A Newton correction,

$$B_N^{\text{new}} := B_N - \frac{\mathcal{R}_N}{1 + A'(L + B_N)}, \quad (6.22)$$

raises the formal order sharply and is useful when one wants high numerical accuracy without storing many explicit blocks.

7 General inversion theory for multiexponential shapes

The Fibonacci function is one instance of a broad class: a dominant exponential multiplied by an analytic germ in smaller exponentials. The same diagonal coefficient mechanism solves the inverse in closed form.

7.1 Analytic-germ theorem

Let $\mathbf{z} = (z_1, \dots, z_m)$, let $\mathbf{n} = (n_1, \dots, n_m) \in \mathbb{N}^m$, and write

$$|\mathbf{n}| := n_1 + \dots + n_m, \quad \mathbf{z}^{\mathbf{n}} := z_1^{n_1} \dots z_m^{n_m}. \quad (7.1)$$

Theorem 7.1 (Inverse of a dominant exponential times an analytic germ). *Let $\alpha > 0$, let $\mu_1, \dots, \mu_m \in \mathbb{C}$ satisfy $\Re \mu_j > 0$, and let Ψ be holomorphic and nonzero near $\mathbf{0}$, with $\Psi(\mathbf{0}) = 1$. Consider a compatible branch of*

$$Y(x) = C e^{\alpha x} \Psi(\zeta_1 e^{-\mu_1 x}, \dots, \zeta_m e^{-\mu_m x}), \quad C \neq 0. \quad (7.2)$$

Set

$$L := \frac{1}{\alpha} \operatorname{Log} \frac{Y}{C}, \quad \xi_j := \zeta_j e^{-\mu_j L}, \quad \Gamma_{\mathbf{n}} := \sum_{j=1}^m n_j \mu_j. \quad (7.3)$$

Then the inverse branch asymptotic to L has the convergent multiexponential expansion

$$x = L + \sum_{\mathbf{n} \in \mathbb{N}^m \setminus \{\mathbf{0}\}} B_{\mathbf{n}} \xi^{\mathbf{n}}, \quad B_{\mathbf{n}} = -\frac{1}{\Gamma_{\mathbf{n}}} [z^{\mathbf{n}}] \Psi(\mathbf{z})^{\Gamma_{\mathbf{n}}/\alpha}. \quad (7.4)$$

The power of Ψ is defined by the local logarithm satisfying $\operatorname{Log} \Psi(\mathbf{0}) = 0$. Since $\Re \Gamma_{\mathbf{n}} > 0$ for $\mathbf{n} \neq \mathbf{0}$, the denominator does not vanish.

Proof. Taking logarithms in equation (7.2) gives

$$L = x + A(x), \quad A(x) = \frac{1}{\alpha} \operatorname{Log} \Psi(\zeta_1 e^{-\mu_1 x}, \dots, \zeta_m e^{-\mu_m x}). \quad (7.5)$$

At $x = L$, the monomial $\xi^{\mathbf{n}}$ is an eigenmonomial of $D = d/dL$ with eigenvalue $-\Gamma_{\mathbf{n}}$. Applying theorem 4.2 gives

$$\begin{aligned} B_{\mathbf{n}} &= -\frac{1}{\Gamma_{\mathbf{n}}} [z^{\mathbf{n}}] \exp(\Gamma_{\mathbf{n}} \alpha^{-1} \operatorname{Log} \Psi(\mathbf{z})) \\ &= -\frac{1}{\Gamma_{\mathbf{n}}} [z^{\mathbf{n}}] \Psi(\mathbf{z})^{\Gamma_{\mathbf{n}}/\alpha}. \end{aligned} \quad (7.6)$$

For convergence, define the analogue of equation (6.1); its derivative with respect to the inverse correction is 1 at the origin. The holomorphic implicit-function theorem supplies a convergent multivariate Taylor series, whose coefficients are uniquely those above. $\square$

Remark 7.2 (One-dimensional inversion, multivariate coefficient extraction). *The inverse variable remains one-dimensional. The several small exponentials serve only as bookkeeping variables for its transseries. This is why the answer requires no multivariate Lagrange determinant: ordinary one-variable Lagrange inversion, combined with diagonal differentiation of exponential monomials, already contains the full multi-index combinatorics.*

7.2 Product-exponential functions

When the analytic germ factors into binomials, every coefficient in equation (7.4) becomes explicit.

Corollary 7.3 (Closed formula for product-exponential shapes). *Under the hypotheses of theorem 7.1, suppose*

$$\Psi(\mathbf{z}) = \prod_{j=1}^m (1 - z_j)^{\beta_j}. \quad (7.7)$$

Then

$$B_{\mathbf{n}} = \frac{(-1)^{|\mathbf{n}|+1}}{\Gamma_{\mathbf{n}}} \prod_{j=1}^m \binom{\beta_j \Gamma_{\mathbf{n}} / \alpha}{n_j}. \quad (7.8)$$

Proof. Raise equation (7.7) to $\Gamma_{\mathbf{n}}/\alpha$, expand each factor by the generalized binomial theorem, and use equation (7.4). $\square$

For the Fibonacci problem, take

$$Y = R = \sqrt{5} \mathcal{F}(x), \quad C = 1, \quad \alpha = \lambda, \quad (\mu_1, \mu_2) = (a + ib, a - ib), \quad (\beta_1, \beta_2) = \left(\frac{1}{2}, \frac{1}{2}\right). \quad (7.9)$$

Then $\Gamma_{r,s}/\alpha = 2\chi_{r,s}$, so $\beta_j \Gamma_{r,s}/\alpha = \chi_{r,s}$, and equation (7.8) reduces exactly to equation (1.7).

7.3 Dominant exponential polynomials

A common source of equation (7.2) is an exponential polynomial with one dominant term. Suppose, for example,

$$Y(x)^p = c_0 e^{\kappa_0 x} \left(1 + \sum_{j=1}^m c_j e^{-\nu_j x} \right), \quad \Re \nu_j > 0. \quad (7.10)$$

On a compatible branch,

$$Y(x) = c_0^{1/p} e^{(\kappa_0/p)x} \left(1 + \sum_{j=1}^m c_j e^{-\nu_j x} \right)^{1/p}, \quad (7.11)$$

so theorem 7.1 applies with

$$\Psi(z) = (1 + z_1 + \cdots + z_m)^{1/p}. \quad (7.12)$$

The Fibonacci square is precisely a three-term exponential polynomial plus a conjugate oscillatory pair, and the factorization equation (3.2) further simplifies its germ into a product.

More generally, any finite exponential sum can be divided by its dominant monomial. Taking a logarithm produces an additive semigroup of smaller exponents, and the diagonal formula applies whether or not the remaining germ factorizes.

8 Formal transseries algebra and block calculus

8.1 A complete exponential-grid algebra

The preceding derivations can be interpreted analytically, but they also make sense formally; this filtered viewpoint is consistent with the grid-based and Hahn-style transseries frameworks in [5, 6, 7]. This matters when the starting expansion is only asymptotic. Fix generators $\mu_1, \dots, \mu_m$ with positive real parts and form the additive monoid

$$\mathfrak{G} := \left\{ \sum_{j=1}^m n_j \mu_j : \mathbf{n} \in \mathbb{N}^m \right\}. \quad (8.1)$$

Because the number of generators is finite and $\Re \mu_j > 0$, only finitely many multi-indices satisfy $\Re(\mathbf{n} \cdot \boldsymbol{\mu}) \leq R$ for a fixed R . After combining any exact exponent collisions, formal sums

$$T(X) = \sum_{\gamma \in \mathfrak{G}} t_\gamma e^{-\gamma X} \quad (8.2)$$

form a complete filtered algebra under the real-part filtration. One may use

$$\text{val } T := \min\{\Re \gamma : t_\gamma \neq 0\} \quad (8.3)$$

for nonzero T . Multiplication adds valuations, while differentiation preserves them:

$$D e^{-\gamma X} = -\gamma e^{-\gamma X}. \quad (8.4)$$

If B has positive valuation, composition is defined by

$$e^{-\gamma(X+B)} = e^{-\gamma X} \exp(-\gamma B). \quad (8.5)$$

Every coefficient below a fixed filtration level receives only finitely many contributions.

Theorem 8.1 (Formal near-identity inversion). *Let A be an exponential-grid series of positive valuation. There is a unique positive-valuation series B satisfying*

$$B(X) = -A(X + B(X)). \quad (8.6)$$

It is given by [equation \(4.2\)](#); equivalently, its coefficients are given by [equation \(4.9\)](#).

Proof. Successive approximation in [equation \(8.6\)](#) determines one filtration layer at a time: replacing B by another series agreeing through valuation ρ changes the right-hand side only at a strictly higher valuation. This proves existence and uniqueness in the complete filtered algebra. Introducing the formal parameter t in the proof of [theorem 4.1](#) is legitimate coefficientwise, because only finitely many products contribute at any fixed filtration level. The same proof therefore yields the stated formulas. $\square$

This theorem separates two issues that are often conflated:

1. formal inversion is controlled by positive valuation and local finiteness of the exponent support;
2. analytic convergence requires additional holomorphy and noncriticality, supplied for the Fibonacci problem by the implicit-function theorem.

8.2 Periodic and quasiperiodic block algebra

Complex conjugate exponents with a common real part are often clearer after grouping them into real phase blocks. Let

$$q = e^{-\rho X}, \quad \boldsymbol{\theta} = \boldsymbol{\theta}_0 + \boldsymbol{\omega} X \pmod{2\pi}, \quad \rho > 0, \quad (8.7)$$

and suppose

$$A(X) = \sum_{n \geq 1} q^n a_n(\boldsymbol{\theta}), \quad (8.8)$$

where the a_n are analytic functions on a torus $\mathbb{T}^d$, or formal Fourier series with appropriate support. Then

$$D = -\rho q \partial_q + \boldsymbol{\omega} \cdot \nabla_{\boldsymbol{\theta}}, \quad (8.9)$$

and in particular

$$D(q^N f(\boldsymbol{\theta})) = q^N (\boldsymbol{\omega} \cdot \nabla_{\boldsymbol{\theta}} - \rho N) f(\boldsymbol{\theta}). \quad (8.10)$$

Theorem 8.2 (Universal block reversion formula). *If*

$$H(X) = X + A(X), \quad H^{-1}(X) = X + \sum_{N \geq 1} q^N b_N(\theta), \quad (8.11)$$

then

$$b_N(\theta) = \sum_{k=1}^N \frac{(-1)^k}{k!} (\omega \cdot \nabla_{\theta} - \rho N)^{k-1} \sum_{\substack{n_1 + \dots + n_k = N \\ n_j \geq 1}} \prod_{j=1}^k a_{n_j}(\theta). \quad (8.12)$$

Proof. The coefficient of q^N in A^k is the inner composition sum in equation (8.12). On a block of order N , every application of D is the fixed differential operator in equation (8.10). Extract the coefficient of q^N from equation (4.2). $\square$

For the Fibonacci inverse, $d = 1$, $\rho = a$, $\omega = b$, and

$$a_n(\theta) = -\frac{2}{an} \cos(n\theta). \quad (8.13)$$

The theorem then reproduces $b_N = P_N/a$. Unlike the closed binomial formula, equation (8.12) applies without factorization and is therefore the natural general-purpose algorithm for log-periodic problems.

If the imaginary parts of the exponent generators have rational rank $d > 1$, the coefficient functions live on $\mathbb{T}^d$ and are quasiperiodic along the one-dimensional orbit $X \mapsto \theta_0 + \omega X$. Commensurate frequencies reduce to an ordinary periodic phase. Incommensurate decay rates should usually be left in the full multiexponential grid rather than forced into a single parameter q .

8.3 Resonances, reality, and absence of small divisors

Two distinct multi-indices can occasionally produce the same exponent Γ_n . Their formal monomials must then be combined before one regards the result as a one-variable exponential series. The coefficient formula remains valid termwise in independent bookkeeping variables and the resonant contributions are subsequently added.

Positive real parts rule out dangerous denominators:

$$|\Gamma_n| \geq \Re \Gamma_n > 0. \quad (8.14)$$

Thus the inversion itself creates no small-divisor problem. For real-valued functions, exponent generators, germ coefficients, and bookkeeping variables should be closed under complex conjugation. The resulting coefficients pair exactly as in equation (4.16).

8.4 Polynomial and logarithmic coefficient extensions

If the input contains factors such as $X^k e^{-\gamma X}$ or $(\log X)^k e^{-\gamma X}$, differentiation is no longer diagonal within a single monomial. Polynomial prefactors lead to finite triangular blocks; logarithms and inverse powers lead to a larger well-ordered differential coefficient algebra in which differentiation is still triangular. The universal formula equation (4.2) remains valid unchanged; only the shortcut equation (4.9) must be replaced by triangular coefficient propagation. This observation is the bridge from pure exponential grids to logarithmic-exponential transseries.

9 Conjugation by a nonlinear dominant skeleton

The leading inverse need not be linear in the logarithm of the target. A simple conjugation reduces a much wider class to the same near-identity problem.

Suppose an inverse equation can be written

$$Z = S(x) + P(x), \quad (9.1)$$

where S is a monotone or locally univalent dominant skeleton with known inverse $T = S^{-1}$, and P is asymptotically small relative to the chosen scale. Set

$$t := S(x), \quad \tilde{A}(t) := P(T(t)). \quad (9.2)$$

Then

$$Z = t + \tilde{A}(t), \quad (9.3)$$

so

$$t = Z + \tilde{B}(Z), \quad \tilde{B} = \sum_{n \geq 1} \frac{(-1)^n}{n!} \frac{d^{n-1}}{dZ^{n-1}} \left(\tilde{A}(Z)^n \right), \quad (9.4)$$

and finally

$$\boxed{x = T(Z + \tilde{B}(Z))}. \quad (9.5)$$

If desired, the outer composition is expanded as

$$x = T(Z) + T'(Z)\tilde{B} + \frac{1}{2}T''(Z)\tilde{B}^2 + \dots. \quad (9.6)$$

This covers dominant skeletons such as $S(x) = \alpha x + \beta \log x$, $S(x) = x \log x$, or a skeleton whose inverse is expressed by a Lambert W -function. The smaller exponential, logarithmic, and oscillatory corrections are then inverted in the conjugated coordinate t . In transseries language, one first inverts the dominant level and only then reverts the well-based perturbation.

Scope of the method

The hypotheses $\Re \mu_j > 0$ and $\Psi(0) = 1$ express genuine decay of the perturbation after the dominant factor has been removed. If some correction has zero real decay, it belongs in the dominant skeleton or in a phase-dependent leading problem, not in the small transseries. Singularities of Ψ , or critical points where the inverse derivative vanishes, determine the analytic convergence boundary even though the formal expansion remains well defined.

10 Algorithms and reproducible coefficient generation

10.1 Four complementary algorithms

The theory suggests four computational routes.

Algorithm A: direct multi-index coefficients

For a factored product-exponential input, enumerate the desired multi-indices $\mathbf{n}$, compute $\Gamma_{\mathbf{n}}$, and evaluate [equation \(7.8\)](#). This is the fastest route for the Fibonacci problem and produces exact coefficients in the parameter τ .

Algorithm B: analytic-germ diagonal extraction

For an unfactored germ Ψ , use equation (7.4). For each $\mathbf{n}$, expand the single variable power $\Psi(z)^{\Gamma_{\mathbf{n}}/\alpha}$ only far enough to extract $z^{\mathbf{n}}$. This avoids constructing irrelevant terms.

Algorithm C: universal block formula

When decay orders are commensurate, represent the perturbation as $A = \sum q^n a_n(\theta)$ and apply equation (8.12). Fourier or polynomial representations of the coefficient functions make the phase differential operator sparse.

Algorithm D: triangular implicit recurrence

Use the fixed-point equation

$$B = -A(X + B) = -\sum_{m \geq 0} \frac{B^m}{m!} D^m A(X). \quad (10.1)$$

At order N , the right-hand side involves only coefficients of B from lower orders, because both A and B have positive valuation. This gives a memory-efficient recursive implementation. Substitution into the original implicit equation provides an immediate residual check.

For the Fibonacci blocks, direct expansion of equation (3.12) is especially convenient. If $W_{<N} = \sum_{n=1}^{N-1} q^n P_n$, then the coefficient of q^N in

$$e^{W_{<N} + q^N P_N} + q^2 e^{-W_{<N} - q^N P_N} - 2q \cos(\theta + \tau(W_{<N} + q^N P_N)) - 1 \quad (10.2)$$

is linear in P_N ; setting it to zero determines the next block.

10.2 Ordering and truncation

For a finite grid with unequal decay rates, total multi-index degree is not always the best notion of order. A robust implementation uses the weight

$$w(\mathbf{n}) := \Re \Gamma_{\mathbf{n}} \quad (10.3)$$

and retains all terms below a prescribed weight. In the Fibonacci problem, all generators have real part a , so $w(r, s) = a(r + s)$ and total degree is exactly the decay order.

When terms are grouped into phase blocks, the remainder should be stated in block norm, as in equation (1.11), rather than by dividing by a possibly vanishing trigonometric polynomial. This convention keeps error statements uniform and numerically meaningful.

10.3 Symbolic verification protocol

A reliable calculation should perform all three checks below:

1. verify conjugation, $C_{s,r} = \overline{C_{r,s}}$, and realness of each grouped block;
2. substitute the truncated W into equation (3.12) and confirm that coefficients through the requested order vanish;
3. compare the truncated inverse against a high-precision numerical root of $\mathcal{F}(x) = y$, preferably at both generic values and exact Fibonacci values.

The Wolfram Language implementation in section B follows this protocol.

11 Numerical validation

For $N \geq 0$, define the truncation

$$X_N(y) := L + \frac{1}{a} \sum_{n=1}^N q^n P_n(\theta; \tau), \quad X_0(y) := L. \quad (11.1)$$

The positive reference root was computed at high precision from $\mathcal{F}(x) = y$, starting from L . The table reports absolute errors. The coefficients through P_6 are given in [sections 1](#) and [A](#).

Table 1: Absolute inverse errors $E_N(y) = |X_N(y) - X(y)|$.

y	exact or numerical $X(y)$	E_0	E_1	E_2	E_4	E_6
2	3	1.12696×10^{-1}	1.52360×10^{-2}	2.96187×10^{-3}	3.76916×10^{-4}	1.28682×10^{-4}
3	4	4.47122×10^{-2}	1.01264×10^{-3}	4.22181×10^{-4}	8.66820×10^{-6}	2.17452×10^{-7}
5	5	1.68278×10^{-2}	2.26345×10^{-4}	1.85965×10^{-5}	2.98868×10^{-8}	1.82331×10^{-11}
10	6.4577930902882116495	5.45185×10^{-4}	1.13522×10^{-5}	1.54252×10^{-7}	3.01709×10^{-11}	6.00018×10^{-15}
100	11.242189770132803894	3.01016×10^{-5}	3.56196×10^{-9}	2.97159×10^{-13}	4.30138×10^{-21}	1.48686×10^{-28}

The first three target values are Fibonacci numbers: $2 = \text{Fib}_3$, $3 = \text{Fib}_4$, and $5 = \text{Fib}_5$, so their exact positive preimages are the integers shown. The table illustrates two features. First, the dominant logarithm L is already useful. Second, retaining the phase-dependent blocks produces the predicted rapid improvement as $q = (5y^2)^{-1}$ decreases. At $y = 100$, six blocks give about twenty-eight decimal digits in the inverse.

11.1 Exact consistency at Fibonacci targets

Binet's formula gives, for $n \geq 1$,

$$\sqrt{5} \text{Fib}_n = \varphi^n (1 - (-1)^n \varphi^{-2n}). \quad (11.2)$$

Hence, when $y = \text{Fib}_n$,

$$L = n + \frac{1}{\lambda} \log (1 - (-1)^n \varphi^{-2n}), \quad (11.3)$$

and the exact inverse correction is

$$X(y) - L = n - L = -\frac{1}{\lambda} \log (1 - (-1)^n \varphi^{-2n}). \quad (11.4)$$

The convergent transseries must sum to this quantity at every sufficiently large Fibonacci target. This supplies a discrete infinite family of exact checks in addition to generic numerical root comparisons.

11.2 Derivative of the inverse

Termwise differentiation of [equation \(1.8\)](#) gives asymptotics for the sensitivity of the inverse. From the first block,

$$\frac{dX}{dL} = 1 - \frac{q}{\lambda} (a \cos \theta + b \sin \theta) + \mathcal{O}(q^2). \quad (11.5)$$

Since $dL/dy = (\lambda y)^{-1}$,

$$\boxed{\frac{dX}{dy} = \frac{1}{\lambda y} \left[1 - q \left(2 \cos \theta + \frac{\pi}{\lambda} \sin \theta \right) + \mathcal{O}(q^2) \right]}. \quad (11.6)$$

This agrees with the implicit identity $X'(y) = 1/\mathcal{F}'(X(y))$ on the outer branch.

12 Structural conclusions and extensions

The calculation reveals several general principles.

Complex exponents are the simplest coordinates. The real oscillation $\cos(\pi x)$ becomes a conjugate pair of decaying exponentials after the dominant term is factored. In those coordinates, reversion is diagonal and the coefficient formula is one line. Trigonometric blocks should be formed only after inversion.

Log-periodicity is inherited from the ratio of exponents. The decay rate is $a = 2 \log \varphi$ and the angular frequency is $b = \pi$. Their ratio $\tau = b/a$ controls every sine coefficient. In a general conjugate pair $\rho \pm i\omega$, the corresponding ratio ω/ρ plays the same role.

A finite factorization can encode infinitely many inverse terms. The exact factorization equation (3.2) contains only two small factors, yet generalized binomial extraction generates the complete two-dimensional coefficient array $C_{r,s}$. No term-by-term matching of the original inverse equation is required.

Formal and analytic conclusions should be separated. Positive valuation proves formal invertibility. Holomorphic implicit-function theory proves convergence. A problem may possess the first property without the second, especially when the supplied direct expansion is merely asymptotic. The same coefficient identities still determine the formal inverse.

Nonlinear leading behavior is handled by conjugation. When the dominant inverse is not a logarithm, one first inverts the skeleton S , then applies near-identity reversion in the skeleton coordinate, and finally composes with S^{-1} . This organizes mixed logarithmic, exponential, and oscillatory scales without changing the core inversion identity.

Further questions. The exact bivariate Taylor series invites a study of its singular variety and therefore of the sharp convergence domain in (ξ, η) . Other natural directions are asymptotics of $C_{r,s}$ along rays in the index lattice, complex continuation of the inverse branches, and automated support management for general Hahn-style transseries with polynomial-logarithmic coefficients.

13 Conclusion

For the real-argument Fibonacci interpolation equation (1.2), the large real inverse is governed by the dominant logarithm $L = \log_\varphi(\sqrt{5}y)$, but its correction is intrinsically log-periodic. The exact answer is the convergent conjugate-pair transseries equation (1.7), with every coefficient given by

$$C_{r,s} = \frac{(-1)^{r+s+1}}{2 \log \varphi} \frac{\binom{r+s+i\tau(r-s)}{r} \binom{r+s+i\tau(r-s)}{s}}{r+s+i\tau(r-s)}.$$

Its real form is a series in $(5y^2)^{-1}$ whose coefficients are finite Fourier polynomials in $\pi \log_\varphi(\sqrt{5}y)$. Total-degree truncation has a uniform algebraic remainder, and the negative inverse branch is simply the negative of the positive one.

The same derivation yields the general diagonal formula equation (7.4) for a dominant exponential multiplied by an analytic germ in smaller exponentials. Product factorizations reduce it to generalized

binomial coefficients; commensurate complex exponents produce periodic blocks; incommensurate frequencies produce torus-valued quasiperiodic blocks; and a nonlinear dominant skeleton is handled by conjugation. These formulas are consequences and reorganizations of classical Lagrange–Bürmann inversion, presented here in a form adapted to exponential and logarithmic transseries. No claim of priority is made for the general classical inversion identities; the contribution of this note is the explicit Fibonacci inverse, its closed coefficient array, and the unified specialization to multiexponential shapes.

A Higher log-periodic blocks

For reference, the next two blocks generated by equation (1.10) are

$$P_5 = \frac{\tau^4 - 95\tau^2 + 240}{6} \cos \theta + \frac{4\tau(31 - 2\tau^2)}{3} \sin \theta \\ + \frac{27\tau^4 - 213\tau^2 + 40}{4} \cos(3\theta) - \frac{7\tau(9\tau^2 - 11)}{2} \sin(3\theta) \\ + \frac{625\tau^4 - 875\tau^2 + 24}{60} \cos(5\theta) - \frac{25\tau(5\tau^2 - 1)}{6} \sin(5\theta), \quad (\text{A.1})$$

$$P_6 = -\frac{200}{3} - \frac{92\tau^4 - 937\tau^2 + 450}{6} \cos(2\theta) - \frac{\tau(8\tau^4 - 418\tau^2 + 1035)}{6} \sin(2\theta) \\ - \frac{4(64\tau^4 - 116\tau^2 + 9)}{3} \cos(4\theta) - \frac{4\tau(64\tau^4 - 620\tau^2 + 261)}{15} \sin(4\theta) \\ - \frac{324\tau^4 - 135\tau^2 + 2}{6} \cos(6\theta) - \frac{\tau(648\tau^4 - 1530\tau^2 + 137)}{30} \sin(6\theta). \quad (\text{A.2})$$

The constant in P_6 agrees with equation (5.6):

$$-\frac{1}{6} \binom{6}{3}^2 = -\frac{200}{3}. \quad (\text{A.3})$$

For exact symbolic computation, an equivalent rising-factorial form of the complex coefficient is often convenient:

$$C_{r,s} = -\frac{(-\chi_{r,s})^{\overline{r}} (-\chi_{r,s})^{\overline{s}}}{a r! s! \chi_{r,s}}, \quad (\text{A.4})$$

where $z^{\overline{n}} = z(z+1) \cdots (z+n-1)$. This avoids gamma-function representations whose removable singularities can obscure the fact that the generalized binomial coefficient is a polynomial for integer lower index.

B Wolfram Language implementation

The following code evaluates the closed transseries directly. It keeps the complex conjugate form internally and takes the real part only after summation.

```
ClearAll["Global`*"];

lam = Log[GoldenRatio];
aa = 2 lam;
tau0 = Pi/aa;

chi[r_Integer?NonNegative, s_Integer?NonNegative] /; r + s > 0 :=
  r + s + I tau0 (r - s);
```

```

cc[r_Integer?NonNegative, s_Integer?NonNegative] /; r + s > 0 :=
  (-1)^(r + s + 1) Binomial[chi[r, s], r] Binomial[chi[r, s], s]/
  (aa chi[r, s]);

ell[y_?NumericQ] /; y > 0 := Log[Sqrt[5] y]/lam;
qq[y_?NumericQ] /; y > 0 := 1/(5 y^2);
theta[y_?NumericQ] /; y > 0 := Pi ell[y];

xApprox[y_?NumericQ, nmax_Integer?NonNegative, wp_: 50] /; y > 0 :=
Module[{yy = N[y, wp]},
  N[Re[
    ell[yy] +
    Sum[
      cc[r, n - r] qq[yy]^n Exp[I (n - 2 r) theta[yy]],
      {n, 1, nmax}, {r, 0, n}
    ], wp]
  ];

f[x_?NumericQ] :=
  Sqrt[2/5] Sqrt[Cosh[2 lam x] - Cos[Pi x]];

xExact[y_?NumericQ, wp_: 80] /; y > 0 :=
Module[{xx, yy = N[y, wp]},
  xx /. FindRoot[
    f[xx] == yy,
    {xx, N[ell[yy], wp]},
    WorkingPrecision -> wp,
    AccuracyGoal -> Floor[wp/2],
    PrecisionGoal -> Floor[wp/2]]
  ];

errorTable[ys_List, orders_List, wp_: 80] :=
Table[
  With[{xe = xExact[y, wp]},
    Prepend[Abs[xApprox[y, #, wp] - xe] & /@ orders, xe]
  ],
  {y, ys}
];

errorTable[{2, 3, 5, 10, 100}, {0, 1, 2, 4, 6}]

```

For exact symbolic Fourier blocks, it is useful to keep τ as an independent symbol and to implement the generalized binomial coefficient as a finite product.

```

ClearAll[gbin, chiT, betaT, p, z, t];

gbin[u_, 0] := 1;
gbin[u_, n_Integer?Positive] :=
  Product[u - k, {k, 0, n - 1}]/n!;

chiT[r_Integer?NonNegative, s_Integer?NonNegative, t_] /;
  r + s > 0 := r + s + I t (r - s);

betaT[r_Integer?NonNegative, s_Integer?NonNegative, t_] /;
  r + s > 0 :=
Module[{ch = chiT[r, s, t]},
  (-1)^(r + s + 1) gbin[ch, r] gbin[ch, s]/ch

```

```

];

p[n_Integer?Positive, z_, t_] :=
FullSimplify[
  TrigReduce@ComplexExpand[
    Sum[
      betaT[r, n - r, t] Exp[I (n - 2 r) z],
      {r, 0, n}
    ],
    Assumptions -> Element[{z, t}, Reals]
  ];

Table[p[n, z, t], {n, 1, 6}]

```

Finally, the real generating equation provides a residual test independent of the complex coefficient construction.

```

ClearAll[wTrunc, blockResidual, qsym];

wTrunc[nmax_Integer?NonNegative, z_, qsym_, t_] :=
  Sum[qsym^n p[n, z, t], {n, 1, nmax}];

blockResidual[nmax_Integer?NonNegative, z_, qsym_, t_] :=
Module[{w = wTrunc[nmax, z, qsym, t]},
  FullSimplify[
    Normal@Series[
      Exp[w] + qsym^2 Exp[-w] -
      2 qsym Cos[z + t w] - 1,
      {qsym, 0, nmax}],
    Assumptions -> Element[{z, t}, Reals]
  ]
];

(* This returns 0: all coefficients through qsym^6 vanish. *)
blockResidual[6, z, qsym, t]

```

C A compact general-purpose derivation template

For reuse on another inverse problem of similar shape, the derivation can be organized as follows.

1. **Factor the dominant term.** Rewrite the direct function as

$$Y(x) = C e^{\alpha x} \Psi(\zeta e^{-\mu x}), \quad \Psi(0) = 1. \quad (\text{C.1})$$

2. **Choose the dominant inverse.** Set

$$L = \alpha^{-1} \text{Log}(Y/C). \quad (\text{C.2})$$

3. **Introduce independent small monomials.** Put

$$\xi_j = \zeta_j e^{-\mu_j L}. \quad (\text{C.3})$$

Keep conjugate monomials independent during coefficient extraction.

4. **Apply the diagonal formula.** For every retained multi-index, calculate

$$B_n = -\Gamma_n^{-1} [z^n] \Psi(z)^{\Gamma_n/\alpha}. \quad (\text{C.4})$$

5. **Group equal decay orders and conjugates.** Convert complex monomials into periodic or quasiperiodic real coefficient blocks only after their coefficients are known.
6. **State the remainder in the correct norm.** If the analytic implicit-function theorem applies, use a uniform block remainder. If the direct expansion is formal, state the result in the corresponding filtered transseries algebra.
7. **Verify.** Substitute the truncation into the exact or truncated defining equation, inspect the first nonzero residual order, and compare with high-precision numerical inversion where available.

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